

Luminosity of object-level credences in II-CG

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1 Pre-meeting notes

- I am now reading more through your thesis and past paper (like Ward et al. (2024)) and thinking about how what we are working on interacts with your previous work
 - One such way is the interaction of “honesty” with your previous definition of “belief”. If we want to contribute to a coherent “Wardian” philosophy of intention, belief, honesty... then maybe our definition of “honesty” should satisfy an adequacy condition of the form “If there is a $c \in \text{dom}(C)$ such that the agent believes $C = c$, then a policy cannot be called ‘honest’ which answers ‘ $\hat{c}$ ’ (for a $\hat{c} \neq c$) to the question ‘What is the value of C ?’.”
 - In particular our current (argmax) definition seems to leave open the possibility that a policy is at once honest and (unintentionally? not sure) deceptive (in the sense of (Ward et al. 2023)). This seems off.
 - * To see how, consider: If I observed that this mushroom was safe to eat, I would eat it. As it stands, I think there is a 60% chance that it is safe and a 40% chance that it is not safe. Since I don’t want to take that 40% chance, I do not eat it. According to (Ward et al. 2023)’s definition, I believe that it is poisonous (right?)
 - * If someone asks me “is the mushroom poisonous?”, and I give my argmax answer (“no”) then I say something which I do not believe, and whose negation I in fact believe.
 - * I am not sure how to analyse the intentionality of this here.
 - But also thinking about whether it would make sense to operationalise honesty just as kind of “the opposite” of deception? I.e. develop the theory along the following lines:
 - * An agent S honestly informs an agent T about ϕ if S intentionally causes T to believe ϕ where S believes ϕ .
 - * An agent S truthfully informs an agent T about ϕ if S intentionally causes T to believe ϕ where ϕ is true.
 - * An agent S honestly/truthfully reports their credence in ϕ to T if S honestly/truthfully informs T about $Cr_S(\phi) = p$
 - Where formulae like $Cr_S(\phi) = p$ is something that needs to be added to the language; and whose truth conditions should be “ $Cr_S(\phi) = p$ iff agent S has credence p in ϕ ”.

- [note: think about whether honesty and truthfulness coincide in the case of credence reporting]
- [also here is a small, pedantic, and potentially completely misguided note on the notational conventions in logic. Feel free to ignore it honestly you use $\phi = \top$ to express ‘ ϕ is true’, right? Is this established notation in causal modelling? Because in philosophical logic you wouldn’t write that; it would be considered an infelicitous mix-up of object language and metalanguage. $\top$ is usually used as an element of the object language, to denote a tautological formula, whereas saying that a formula is true is in general only possible in the meta-language. Also propositions can be equivalent ($\Leftrightarrow$) but not “identical” ($=$); terms can be identical (as in $d = c$), but $\top$ is not a term (but a sentence). To say of a formula ϕ that $\phi \Leftrightarrow \top$ would be to say that it is a tautology. Truth cannot be a predicate in the object language of a sufficiently strong theory (anything that can do Robinson arithmetic at least, because then you get Gödel numbering and therefore self-referentiality, which interacts problematically with a truth-predicate). In the object-language, we would just write ϕ if ϕ is the case. When you write e.g. ‘agent i observes that ϕ is true’, I think you can just say ‘agent i observes ϕ ’ and write for the corresponding decision e.g. D_ϕ^i , and for the decision under the observation that ϕ is false $D_{\neg\phi}^i$, in the meta-language you would write $\models \phi$ if the system asserts that ϕ is true; and if you want to say something like ‘in setting $E = e$ and policy profile $\square$, $X = x$ is true’, you might write $X(\square, e) \models X = x$]
- * An agent S honestly/truthfully reports on their best guess for the value of variable C to T if S honestly/truthfully informs T about the member of the set $\{c = \operatorname{argmax}_{c \in \operatorname{dom}(C)} Cr_S(C = c \mid \operatorname{Pa}^D)\}$ which S actually believes.
- * A question from agent T to agent S about formula ϕ is a request for S to honestly inform T about ϕ .
- * A question from agent T to agent S about their credence in formula ϕ is a request for S to honestly report their credence to T .
- * A question from agent T to agent S about their best guess for variable C is a request for S to honestly report their best guess for the value of C to T .
- * An agent A is capable in environment O if $\mathcal{M}^A$ is equivalent to $\mathcal{M}^O$ up to the utility function.
- * An agent is capably honest if the agent has adopted a honest policy and is capable
- * For an environment O containing a question for agent A about formula ϕ at decision node D say that O , say that the agent is in a position to know ϕ at D if $Pr_{\mathcal{M}^O}(\phi \mid \operatorname{Pa}^D) = 1 \Leftrightarrow \phi$.
- * Theorem: A capably honest agent who is in a position to know ϕ is truthful.

References

- Ward, Francis Rhys, Francesco Belardinelli, Francesca Toni, and Tom Everitt. 2023. “Honesty Is the Best Policy: Defining and Mitigating AI Deception.” 2023. <https://doi.org/10.48550/ARXIV.2312.01350>.
- Ward, Francis Rhys, Matt MacDermott, Francesco Belardinelli, Francesca Toni, and Tom Everitt. 2024. “The Reasons That Agents Act: Intention and Instrumental Goals.” February 15, 2024. <https://doi.org/10.48550/arXiv.2402.07221>.